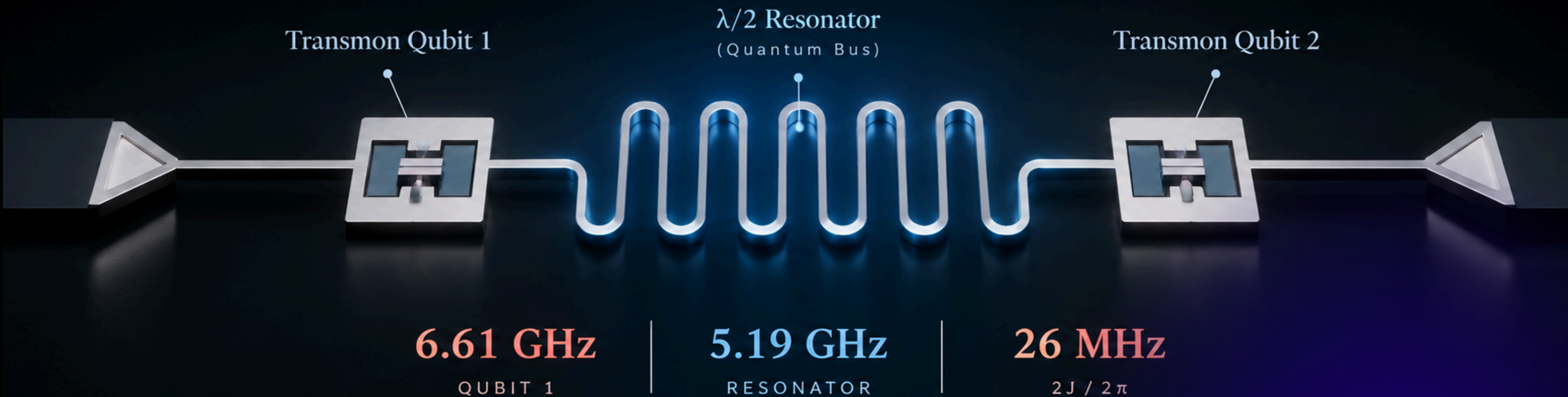


# Can Geometry Reproduce Quantum Behavior?

ENGINEERING A TWO-QUBIT SUPERCONDUCTING CIRCUIT



# From Experimental Physics *to* Engineering Targets

WHAT DOES "REPRODUCE" MEAN?

*Same physics.  
New design.*

**Reference:** Majer et al., *Nature* 2007

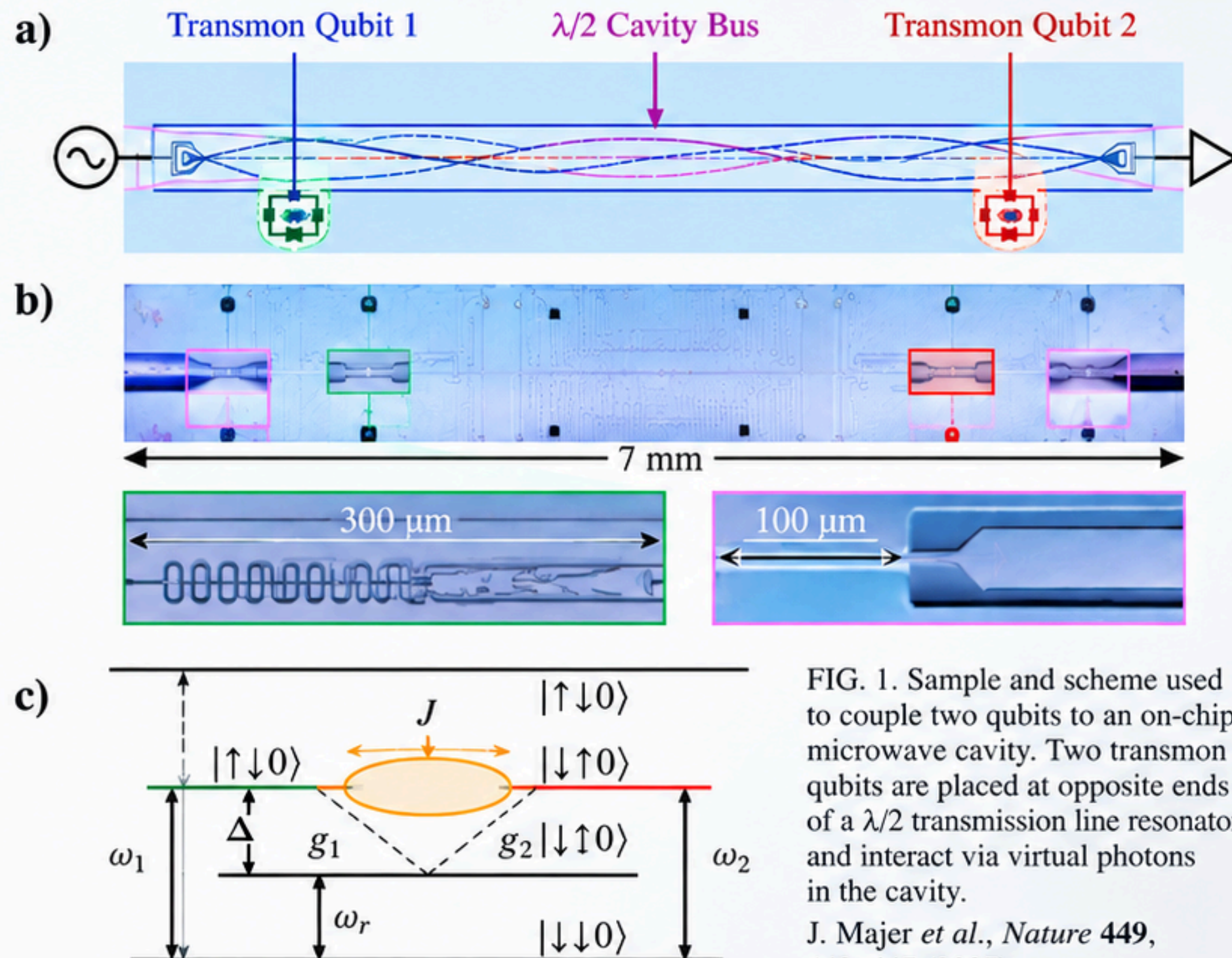


FIG. 1. Sample and scheme used to couple two qubits to an on-chip microwave cavity. Two transmon qubits are placed at opposite ends of a  $\lambda/2$  transmission line resonator and interact via virtual photons in the cavity.  
J. Majer et al., *Nature* **449**, 443–447 (2007).

## Target Parameters from the Reference

THE PHYSICAL BEHAVIOR WE AIM TO REPRODUCE



**QUBITS (Transmons)**

$$f_{q1} = 6.61 \text{ GHz}$$

$$f_{q2} = 6.52 \text{ GHz}$$

In this hackathon, we use fixed-frequency transmons (the flux-tunability via SQUID in the experiment is not included in our model).



**RESONATOR ( $\lambda/2$ )**

$$f_r = 5.19 \text{ GHz}$$

A half-wave transmission line resonator acts as a quantum bus, mediating the interaction between qubits.



**INTERACTION**

$$2J/2\pi = 26 \text{ MHz}$$

An effective qubit-qubit coupling via virtual photons in the cavity, observed as an avoided crossing in spectroscopy.

ADDITIONAL TARGETS

Anharmonicities

$$\alpha_1 = 424 \text{ MHz}$$

$$\alpha_2 = 442 \text{ MHz}$$

Dispersive shifts

$$|\chi_1| = 5.9 \text{ MHz}$$

$$|\chi_2| = 7.4 \text{ MHz}$$

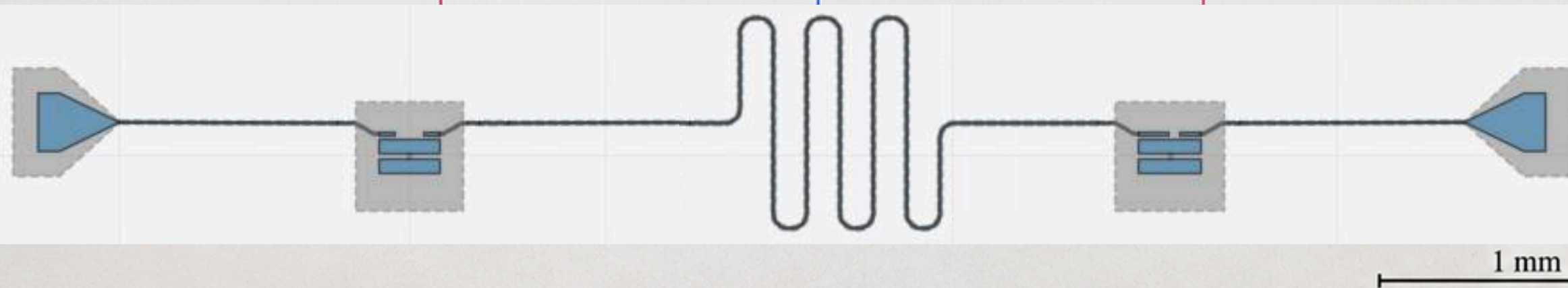
# From Geometry *to* Quantum Parameters

WHICH GEOMETRICAL KNOBS CONTROL THE PHYSICS?

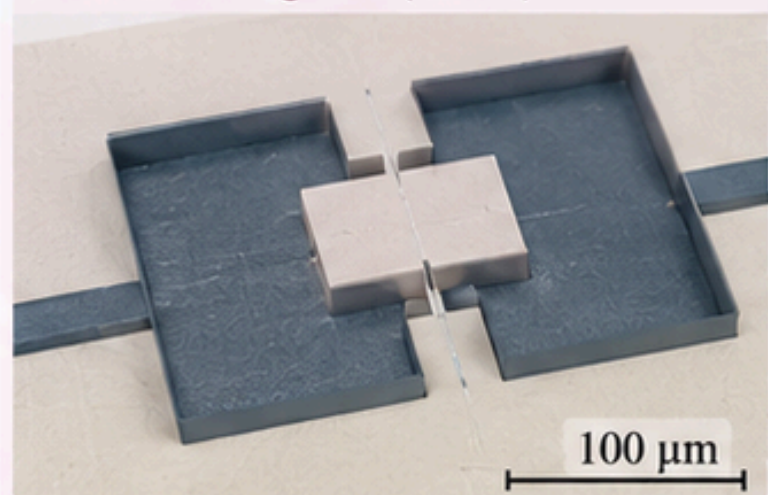
*Design the physics.*

Top view (Qiskit Metal design)

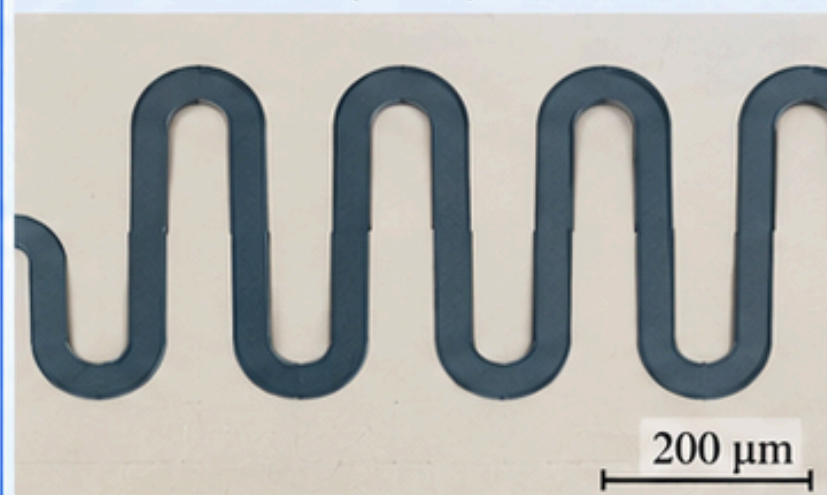
Transmon Qubit 1       $\lambda/2$  Resonator (Quantum Bus)      Transmon Qubit 2



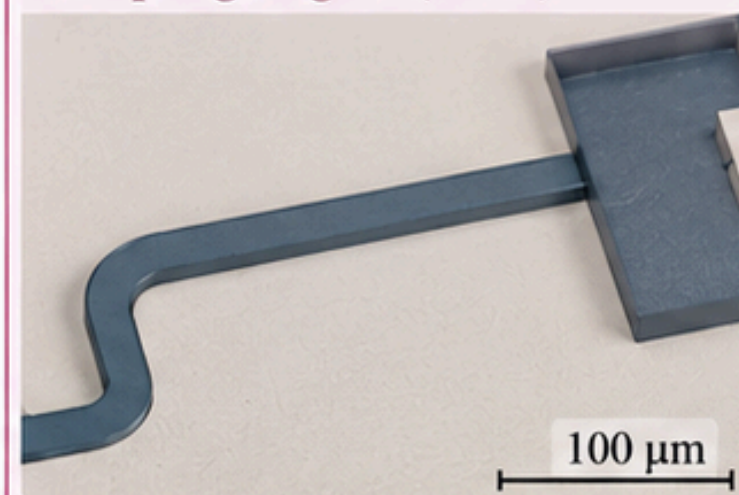
Transmon Qubit (zoom)



$\lambda/2$  Resonator (zoom)



Coupling Region (zoom)



## Key Design Parameters

AND THEIR PHYSICAL IMPACT

### Qubit Pad Geometry

- Pad width / length
- Gap (JJ)
- Island size



Qubit frequency

$$f_q$$

Anharmonicity

$$\alpha$$

### Resonator Geometry

- Total length  $L_{eff}$
- Line width
- Meander spacing



Resonator frequency

$$f_r$$

Mode properties

### Coupling Geometry

- Coupling gap  $g$
- Coupling line length
- Coupling to resonator



Coupling strength

$$J$$

Dispersive shifts

$$\chi$$

**Our Workflow**

Design geometry  
(Qiskit Metal)



Simulate EM response  
(HFSS)



Extract mode parameters  
(EPR analysis)



Compare with targets  
 $f_q, f_r, \alpha, \chi, J$

*Geometry informs physics.*

# From Design *to* Electromagnetic Response

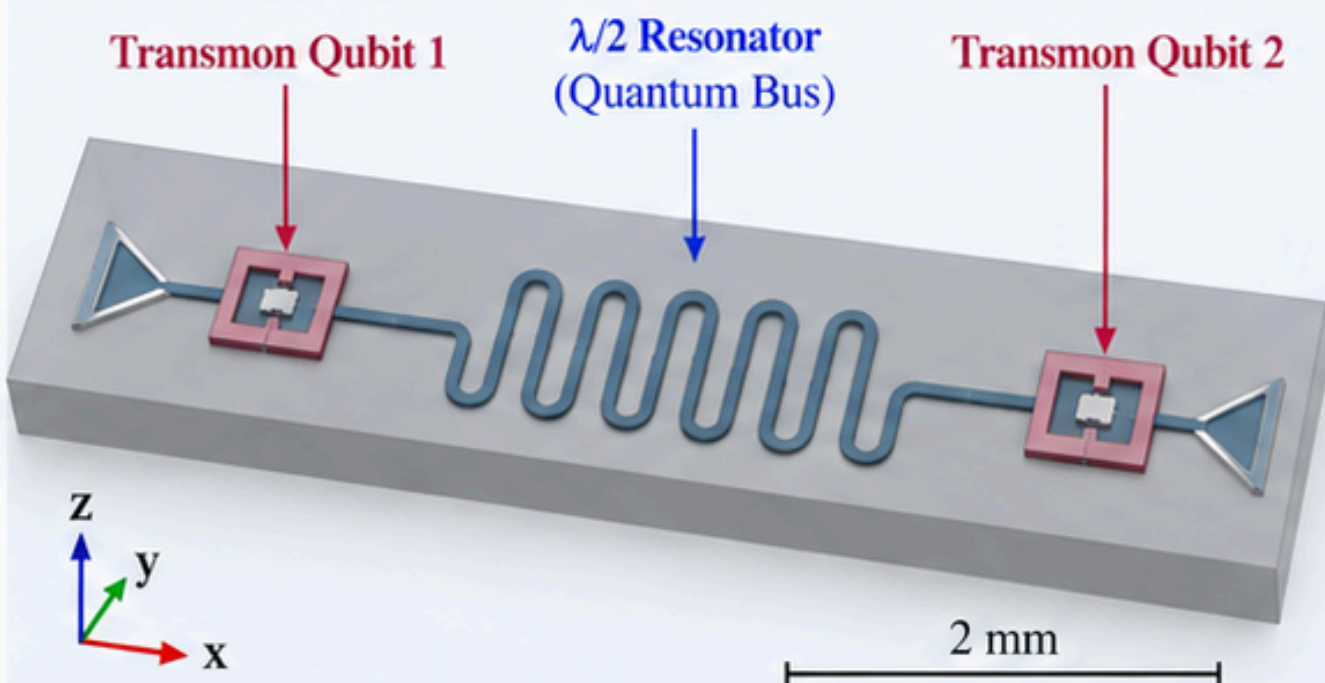
DOES OUR GEOMETRY SUPPORT THE TARGET MODES?

*Same design.  
Real physics.*

## 1 Simulation Setup

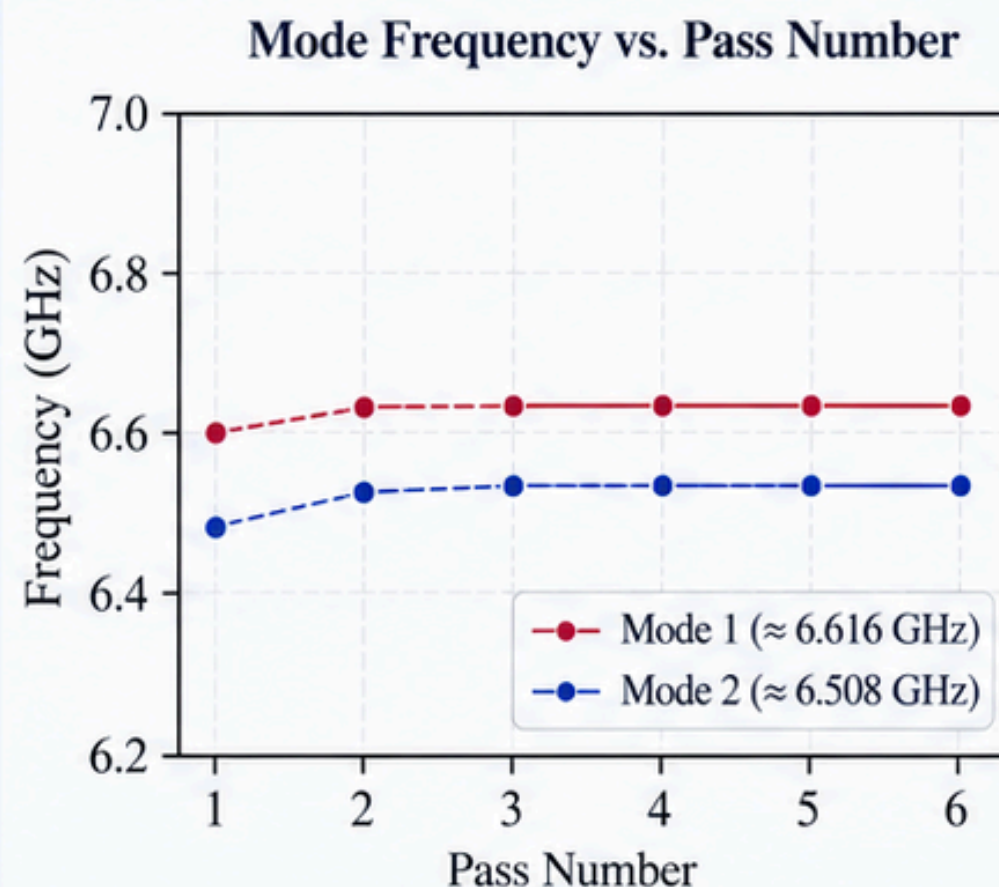
FROM LAYOUT TO HFSS

- Geometry designed in Qiskit Metal and exported to Ansys HFSS (3D).
- Superconducting aluminum ( $t = 10\text{--}20\text{ nm}$ ) on Si substrate ( $\epsilon_r = 11.45$ ).
- Open boundary (radiation) with vacuum box.
- Eigenmode analysis with adaptive mesh refinement.



## 2 Convergence Analysis

ADAPTIVE MESH REFINEMENT



Frequencies converge with mesh refinement, indicating a stable and reliable solution.

## 3 Extracted Eigenfrequencies

FINAL VALUES FROM HFSS

| Mode   | Simulated (GHz) | Target (GHz) | Error (%) |
|--------|-----------------|--------------|-----------|
| Mode 1 | 6.616054        | 6.61         | 0.09      |
| Mode 2 | 6.507715        | 6.52         | 0.19      |



The simulated frequencies are in excellent agreement with the target values from the reference.

## 4 Next Step

MODE IDENTIFICATION

- Frequency matching alone is not sufficient.
- In the next step, we analyze the field distributions and participation ratios to confirm the physical nature of each mode (qubit modes vs. resonator mode).
- This ensures that the extracted frequencies correspond to the intended quantum elements.

# Frequency Match Is *Not* Enough

FROM NUMERICAL AGREEMENT TO PHYSICAL VALIDATION

*Same numbers  
are not enough.*

## 1 Simulated Eigenfrequencies

FINAL VALUES FROM HFSS

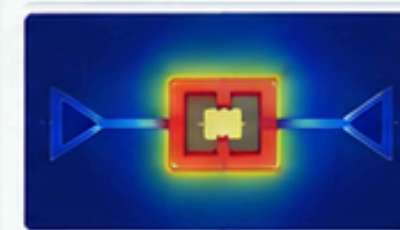
| Mode   | Simulated (GHz) | Target (GHz) | Error (%) |
|--------|-----------------|--------------|-----------|
| Mode 3 | 6.616054        | 6.61         | 0.09      |
| Mode 2 | 6.507715        | 6.52         | 0.19      |
| Mode 1 | 5.105422        | 5.19         | 1.62      |

## 2 But Which Mode Is Which?

FREQUENCY MATCH  $\neq$  PHYSICAL IDENTITY6.616054 GHz  $\xrightarrow{?}$  Qubit 1 ?6.507715 GHz  $\xrightarrow{?}$  Qubit 2 ?Another mode  
(at higher frequency?)  $\xrightarrow{?}$  Resonator ?  
( $\lambda/2$  Bus)**Numerical closeness alone does not  
prove the physical nature of each mode.**

## 3 Validation Path

FROM FIELDS TO QUANTUM PARAMETERS



### Field Distribution

Where is the electromagnetic energy localized?



### Energy Participation Ratio (EPR)

How strongly does each element participate in each mode?

 $\alpha$   $\chi$   $g$   $J$ 

### Quantum Parameters

Extract anharmonicity, dispersive shift, and coupling strengths.



**Next: Validate the physical identity of each mode** using field distributions and EPR analysis,  
and then extract the key quantum parameters.

*From frequencies,  
to physics.*

# Engineering the Geometry

NOT TRIAL AND ERROR — CONTROLLED TUNING

*Same physics.  
Smarter design.*

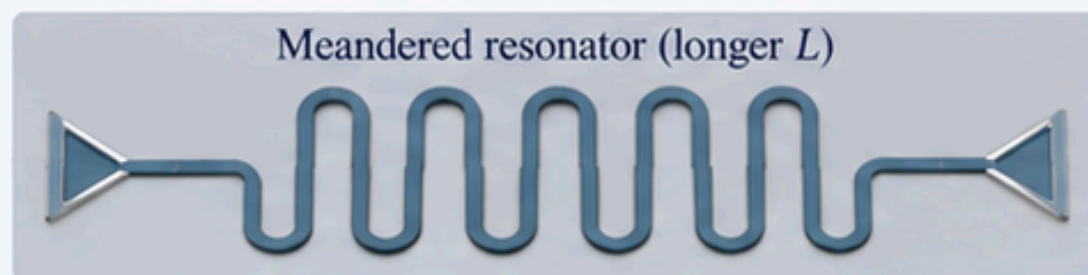
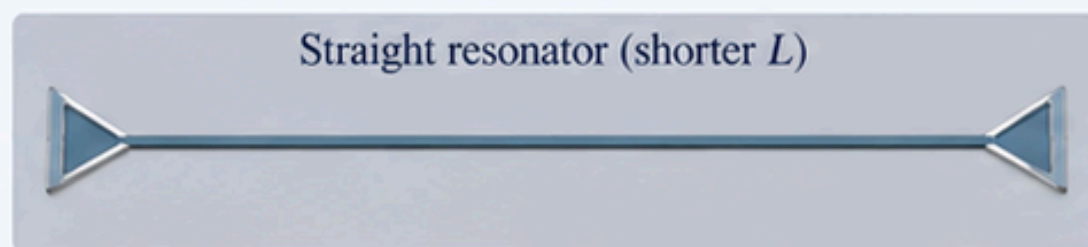
1

## Resonator Geometry ( $L$ )

CONTROLS THE RESONATOR FREQUENCY

- The resonator frequency is mainly set by the effective electrical length  $L$ .
- We use a meandered structure to achieve the required length in a compact footprint.

$$f_r \approx \frac{1}{2L\sqrt{L_k C}} \quad \left| \begin{array}{l} \text{Increase } L \\ \rightarrow \text{Decrease } f_r \end{array} \right.$$



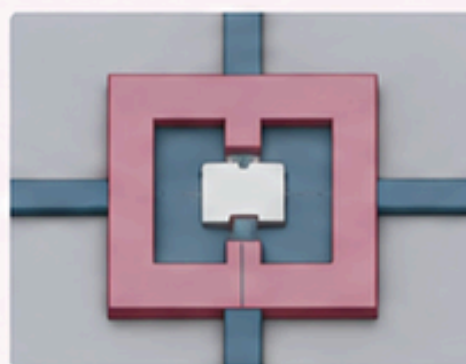
2

## Transmon Geometry ( $C_\Sigma$ )

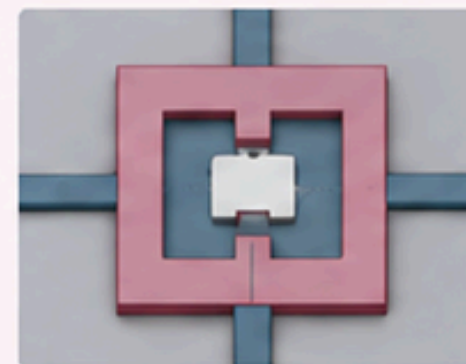
CONTROLS QUBIT FREQUENCY AND ANHARMONICITY

- The pad size and pocket geometry change the total capacitance  $C_\Sigma$ .
- This directly affects the charging energy  $E_C$ , the qubit frequency, and the anharmonicity.

$$f_q \approx \frac{1}{2\pi\sqrt{L_J C_\Sigma}} \quad E_C = \frac{e^2}{2C_\Sigma} \quad \left| \begin{array}{l} \text{Increase } C_\Sigma \\ \rightarrow \text{Decrease } f_q \\ \rightarrow \text{Decrease } E_C \\ \rightarrow \text{Smaller } |\alpha| \end{array} \right.$$



Qubit 1 geometry  
(pad width = 363  $\mu\text{m}$ )



Qubit 2 geometry  
(pad width = 367  $\mu\text{m}$ )

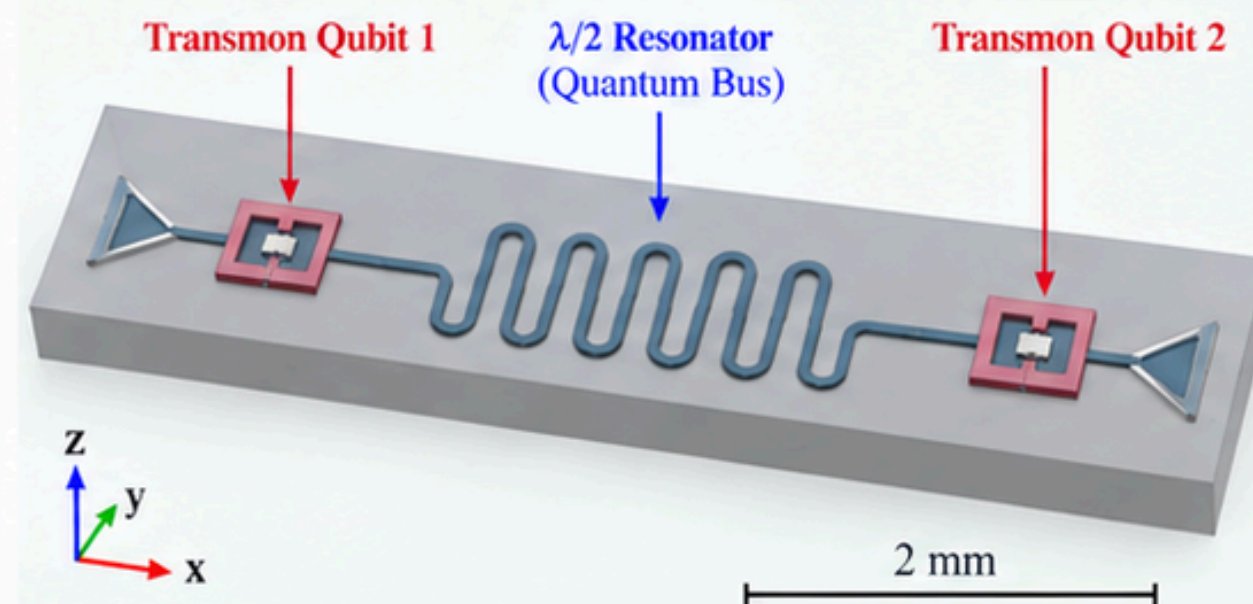
3

## Coupling Geometry ( $C_g$ )

CONTROLS MODE INTERACTION

- The coupling region (distance, overlap, and trace width) sets the coupling capacitance  $C_g$ .
- This determines the interaction strength  $g$  and influences the dispersive shift  $\chi$  and effective coupling  $J$ .

$$g \propto C_g \quad \left| \begin{array}{l} \text{Increase } C_g \\ \rightarrow \text{Stronger coupling } (g, J) \\ \rightarrow \text{Larger dispersive shift } (\chi) \end{array} \right.$$



We tuned the geometry according to *physics* — not numbers according to a solver.

*From design choices,  
to target performance.*

# Can Geometry Determine Quantum Behavior?

FROM DESIGN DECISIONS TO PHYSICAL INSIGHTS

*Geometry  
is part of the physics.*

1

## What We Achieved

A PHYSICS-DRIVEN DESIGN AND VALIDATION

- ✓ Designed a two-transmon circuit with a  $\lambda/2$  resonator in Qiskit Metal.
- ✓ Performed full-wave electromagnetic simulation in Ansys HFSS with adaptive mesh refinement.
- ✓ Achieved excellent agreement with the target frequencies:

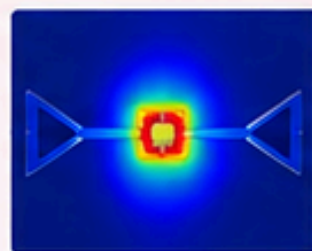
| Mode   | Simulated (GHz) | Target (GHz) | Error (%) |
|--------|-----------------|--------------|-----------|
| Mode 1 | 6.616054        | 6.61         | 0.09      |
| Mode 2 | 6.507715        | 6.52         | 0.19      |

- ✓ Established a clear relation between geometric parameters and physical properties ( $f_r, f_q, \alpha, g, \chi, J$ ).

2

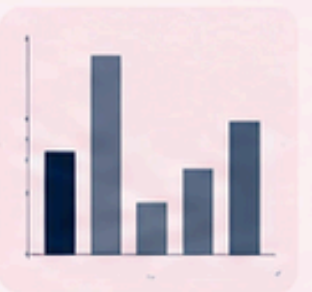
## What Remains

FROM EM MODES TO QUANTUM PARAMETERS



### Physical mode identification

Confirm which mode corresponds to each physical element using field distributions and energy participation ratios (EPR).



### Extraction of quantum parameters

Compute anharmonicity ( $\alpha$ ), dispersive shift ( $\chi$ ), coupling strength ( $g$ ), and effective coupling ( $J$ ) using EPR.



### Further validation

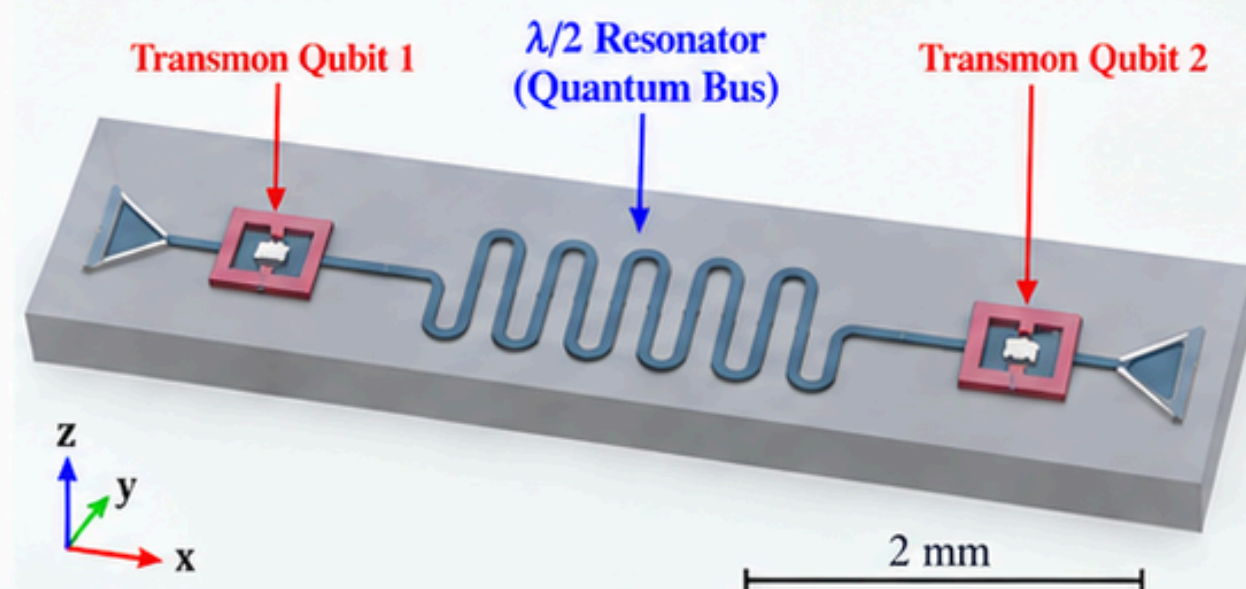
Include substrate and fabrication effects, and validate robustness against design variations.

3

## Key Takeaways

LESSONS BEYOND THE NUMBERS

- ✓ **Geometry** is a powerful design tool, not just a drawing.
- ✓ Physical insight leads to **efficient design and faster convergence**.
- ✓ Numerical agreement is necessary but not sufficient — **field analysis** and **EPR** are essential.
- ✓ A **compact, integrated design** can meet the required specifications for superconducting quantum circuits.



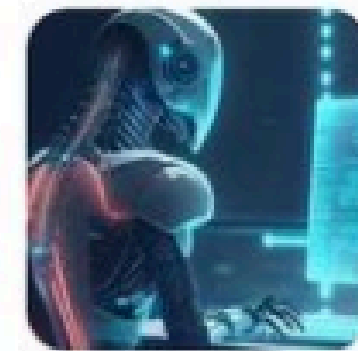
**Geometry → Electromagnetics → Quantum Behavior**

We didn't copy the circuit. We engineered its physics.

*Same physics.  
Smarter design.*

# alirezaarab171/**Qiskit-Metal-HFSS**

Qiskit Metal and HFSS simulations



1

Contributor



0

Issues



0

Stars



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Forks



## **alirezaarab171/Qiskit-Metal-HFSS: Qiskit Metal and HFSS simulations**

Qiskit Metal and HFSS simulations. Contribute to alirezaarab171/Qiskit-Metal-HFSS development by creating an account on GitHub.



GitHub

J. Majer, J. M. Chow, J. M. Gambetta, J. Koch, B. R. Johnson, J. A. Schreier, L. Frunzio, D. I. Schuster, A. A. Houck, A. Wallraff, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, "Coupling superconducting qubits via a cavity bus," *Nature*, vol. 449, no. 7161, pp. 443–447, 2007.